



**IILM University**

# **School of Computer Science and Engineering SUMMER INTERNSHIP PRESENTATION**

## **AI Automation and Full-Stack Software Development**

**HARSHIT ANAND**

**Enrollment No.: CS-23411012**

**B.Tech CSE | 7th Semester | 4CSE1**

**Organization: ScaleOn**



# Internship Overview

## **AI Automation**

Summer Internship focused on practical AI automation and full-stack software development.

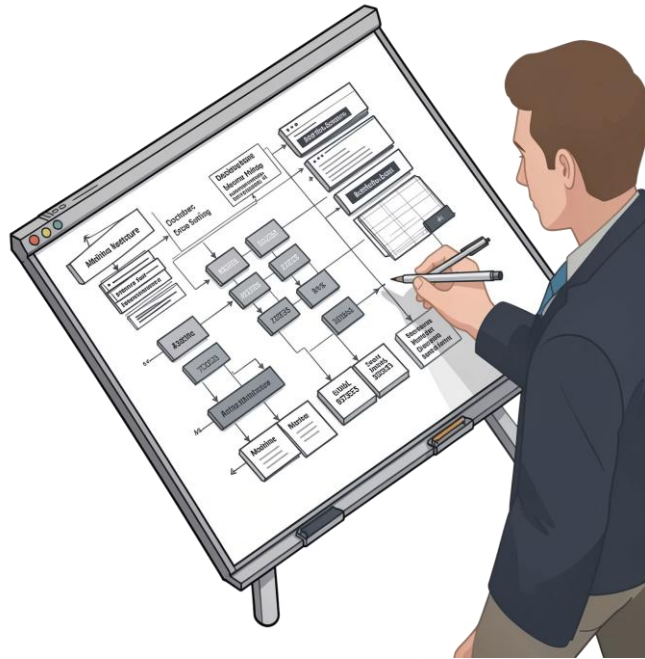
## **The Major Project**

LinkedIn Integration Module.

## **Hands-On Exposure**

APIs, OAuth 2.0, AI services, dashboards, databases, authentication and GitHub collaboration.

# Objectives



## 1 Real-World Solutions

Build practical software solutions for real-world automation and platform-management problems.

## 2 Secure Integrations

Integrate external services securely through APIs and OAuth.

## 3 Generative AI

Use generative AI to automate email classification and response drafting.

## 4 Full-Stack Platform

Contribute to a full-stack internship management and learning platform.

# Project — LinkedIn Integration Module

## FastAPI Backend

FastAPI backend for LinkedIn account integration and social-media publishing.

## OAuth 2.0 Flow

OAuth 2.0 connection flow without handling user passwords.

## Core Features

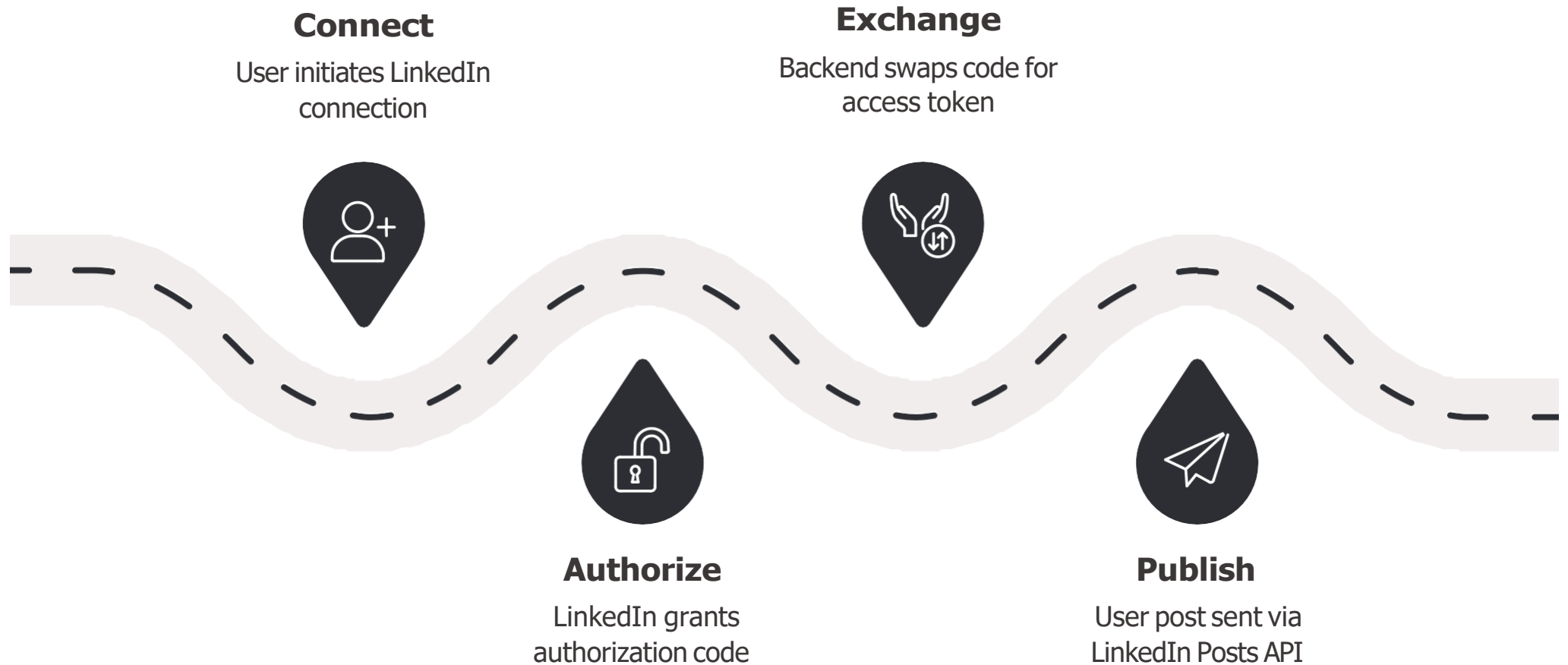
Profile retrieval, text posting, image posting and disconnect functionality.

## Tech Stack

FastAPI, httpx, SQLAlchemy/SQLite, python-dotenv and Docker.



# LinkedIn — Architecture & Workflow



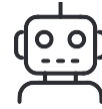
The backend exchanges the authorization code for an access token, stores the connected profile and token, then uses the saved token to publish posts directly to LinkedIn.

# Technologies & Tools



## Backend & APIs

Python, FastAPI, Uvicorn, httpx, SQLAlchemy, SQLite, Docker.



## AI & Data

Streamlit, Plotly, Google Gemini API, IMAP/SMTP, CSV logging.



## Frontend

Next.js, React, TypeScript, Tailwind CSS, Node.js, Express.js.



## Database & UI

PostgreSQL, Prisma ORM, JWT, Framer Motion and Lucide React.

# Learning Outcomes & Challenges

## Learning

OAuth 2.0, generative AI integration, REST APIs, authentication, databases and layered architecture.

## Challenges Solved

API constraints, credential security, duplicate processing, human control in AI automation, complex workflows and team collaboration.

